Asymmetry effects in untrusted noise securuty analysis

April 29, 2026

1 Imperfect gaussian measurements

We begin with a brief discussion of asymmetrical coherent measurements, schematically depicted in Fig. 1. Asymmetry effects are induced by imbalance (unequal transmission and reflection coefficients) of the beamsplitters and unequal quantum efficiency of the detectors. For detailed derivations of the expressions in this section see Ref. [1].

Q symbol of the POVM describing noisy homodyne measurements (see Fig. 1a) is given by

$$\mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_G}} \exp\left[-\frac{(x-\langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \tag{1}$$

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \phi = \arg\alpha_L, \tag{2}$$

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \tag{3}$$

where α (α_L) is the complex amplitude of signal (LO) mode, $\hat{x}_\phi$ is the phase-rotated quadrature operator; x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1+\eta_2)CS\,|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1+\eta_2)\,CS}\,|\alpha_L| \tag{4}$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1+\eta_2)CS]^2}. \tag{5}$$

By computing inverse Forier transform of Q symbol (1), we obtain anti-normal ordered characteristic function [2]. From this characteristic function we can deduce the covariance matrix of the noisy homodyne measurements as follows

$$\Sigma_m^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}+\sigma_N, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{6}$$

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{7}$$

where $\Sigma_{\mathrm{id}}^{(\mathrm{H})}$ is the covariance matrix of sharp measurement, $R(\phi) = \begin{pmatrix}\cos\phi & -\sin\phi \\ \sin\phi & \cos\phi\end{pmatrix}$ is the rotation matrix; σ_N is the excess noise variance that accounts for asymmetry effects is given by

$$\sigma_N = \sigma_x - 1. \tag{8}$$

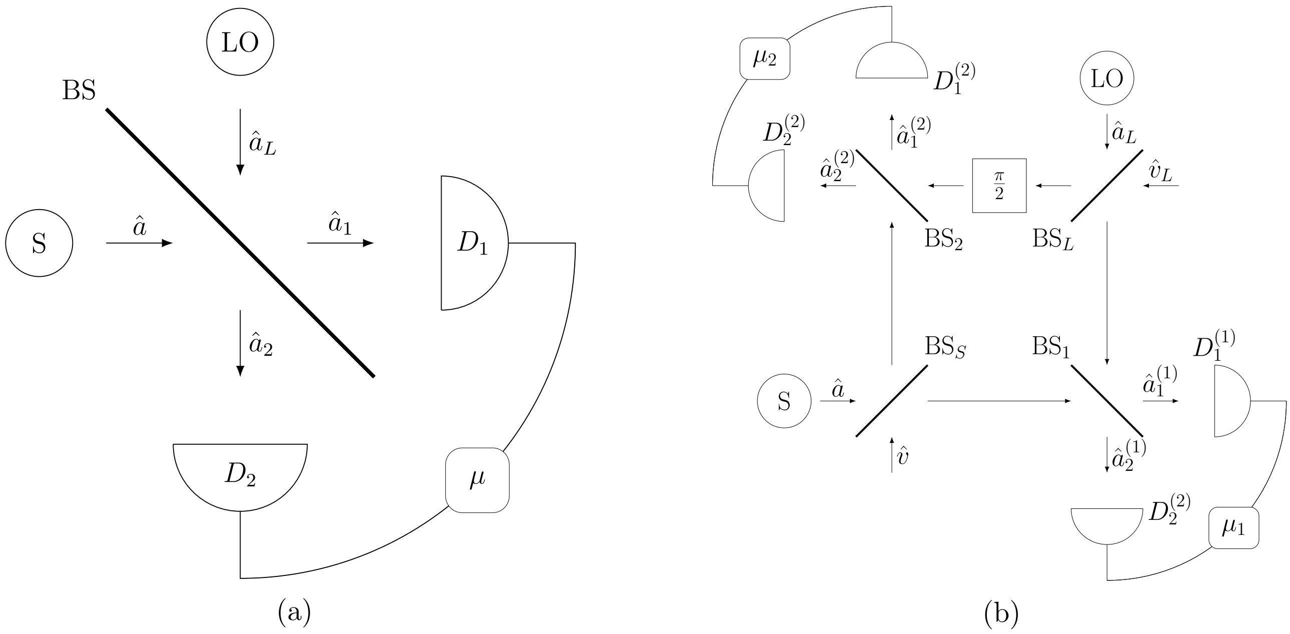


Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beamsplitter with transmission and reflection coefficients C and S, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

For double homodyne detection, Q symbol is given by

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \exp\left[-\frac{(x_1 - \operatorname{Re}\alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \operatorname{Im}\alpha e^{-i\phi})^2}{\sigma_2}\right], \tag{9}$$

$$\sigma_G^{(i)} = (\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2)|\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L \alpha_L, \quad \alpha_L^{(2)} = -i S_L \alpha_L, \tag{10}$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)}) C_1 S_1 C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)} S_1^2 - \eta_2^{(1)} C_1^2)|\alpha_L^{(1)}| \right\}, \tag{11}$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)}) C_2 S_2 S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)} S_2^2 - \eta_2^{(2)} C_2^2)|\alpha_L^{(2)}| \right\}, \tag{12}$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{13}$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2} \tag{14}$$

give the quadrature variances σ_1 and σ_2.

Following the same procedure used to derive Eq. (6), we obtain the covariance matrix of the double homodyne POVM,

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1, \delta_2)R^{\mathrm{T}}(\phi), \tag{15}$$

$$\delta_i = 2\sigma_i - 1. \tag{16}$$

The POVM that describes sharp double homodyne measurements uses the set of squeezed states,

$$|z, r, \phi\rangle = \hat{R}(\phi)\hat{D}(z)\hat{S}(r)|0\rangle = \hat{D}(ze^{i\phi})\hat{S}(re^{2i\phi})|0\rangle, \tag{17}$$

where z is the coherent state amplitude, r is real-valued squeezing parameter, and ϕ is squeezing phase, $\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}$ is the displacement operator, and $\hat{S}(r) = e^{\frac{1}{2}\left(r\hat{a}^{\dagger 2} - r^*\hat{a}^2\right)}$ is the squeezing operator.

The covariance matrix of squeezed states reads

$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi)\begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^{\mathrm{T}}(\phi). \tag{18}$$

An important point is that, for the difference between comvariance matrices of noisy and sharp double homodyne measurements, given by Eq. (15) and Eq. (18) respectively, is positive semi-definite, the squeezing parameter r must be ranged in the interval defined by parameters of the scheme (see Ref. [1])

$$r \in [r_2, r_1]\,,\; r_1 = \ln\delta_1^{1/2},\; r_2 = \ln\delta_2^{-1/2}. \tag{19}$$

2 Gaussian channels modeling imperfect measurements

For arbitrary state $\hat{\rho}$ and POVM $\hat{\Pi}^{(x)}$, the outcome probability is

$$p(x|\hat{\rho}) = \mathrm{Tr}\left(\hat{\rho}\,\hat{\Pi}^{(x)}\right). \tag{20}$$

The measurement statistics of Gaussian states with noisy Gaussian measurements can be calculated by applying the noise map $\mathcal{C}$ to ρ_G, or the dual map $\mathcal{C}^*$ to the ideal POVM $\hat{\Pi}_{\mathrm{id}}$ [2], i.e.

$$p(x|\hat{\rho}_G) = \mathrm{Tr}\left\{\mathcal{C}(\hat{\rho}_G)\,\hat{\Pi}_{\mathrm{id}}^{(x)}\right\} = \mathrm{Tr}\left\{\hat{\rho}_G\,\mathcal{C}^*(\Pi_{\mathrm{id}}^{(x)})\right\}. \tag{21}$$

An important property is that the same measurement statistics can be described by different Gaussian channels $\mathcal{C}$, provided their dual maps $\mathcal{C}^*$ yield the same effective POVM $\hat{\Pi}_G^{(x)} = \mathcal{C}^*\left(\hat{\Pi}_{\mathrm{id}}^{(x)}\right)$.

Note that, as measurements are clarified to be Gaussian, the channels $\mathcal{C}$ and $\mathcal{C}^*$ will be Gaussian as well. It follows that the channel $\mathcal{C}$ is fully characterized by matrices $X_{\mathcal{C}}$ and $Y_{\mathcal{C}}$, transforming the first and second moments of Gaussian state ρ_G, $\boldsymbol{r}_G$ and Σ_G respectively, as follows

$$\begin{aligned} \boldsymbol{r}_G &\mapsto X_{\mathcal{C}}\boldsymbol{r}_G, \\ \Sigma_G &\mapsto X_{\mathcal{C}}\Sigma_G X_{\mathcal{C}}^{\mathrm{T}} + Y, \end{aligned} \tag{22}$$

Matrices charactarizing the dual channel $\mathcal{C}^*$ are given by

$$
\begin{aligned}
X_{\mathcal{C}^*} &= X_{\mathcal{C}}^{-1}, \\
X_{\mathcal{C}^*} &= X_{\mathcal{C}}^{-1} Y_{\mathcal{C}} X_{\mathcal{C}}^{-1\,\mathrm{T}},
\end{aligned} \tag{23}
$$

and act on $\boldsymbol{r}_G$ and Σ_G in the same fashion as Eq. (22).

In what follows, we will consider two distinct quantum channels to describe the imperfect measurements, namely attenuator channel and additive noise channel. In accordance with Eq. (21), channel $\mathcal{C}$ will be defined in a way that its' dual channel $\mathcal{C}^*$ will transform the covariance matrix of ideal measurement Σ_{id} into the covariance matrix of noisy Gaussian measurement Σ_m,

$$
\mathcal{C}^* : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} \tag{24}
$$

2.1 Attenuator channel

First, we shall use the attenuator channel $\mathcal{E}_N^{(\gamma)}$, characterized by matrices [2]

$$
X^{(\gamma;\,\mathcal{E}_N^{(\gamma)})} = \cos\theta_\gamma \mathbb{1}, \quad Y^{(\gamma;\,\mathcal{E}_N^{(\gamma)})} = \sin^2\theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \tag{25}
$$

where Σ_{env} is the covariance matrix of the state decribing the enviroment. This channel acts on first and second moments in accordance with Eq. (22).

Using Eq. (24) together with Eq. (22), we construct the dual channel $\mathcal{E}_N^{*(\gamma)}$ so that it transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing the covariance matrix $\Sigma_{\text{id}}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows

$$
\mathcal{E}_N^{*(\gamma)} : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta_\gamma} \Sigma_{\text{id}}^{(\gamma)} + \tan^2\theta_\gamma \Sigma_{\text{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{26}
$$

Now we derive explicit experessions coefficients of (26).

First, we will consider homodyne detection. If the enviroment is decribed by vacuum, i.e.

$$
\Sigma_{\text{env}}^{(\mathrm{H})} = \mathbb{1}, \tag{27}
$$

from covariance matrices of ideal (7) and noisy (6) measurements, we have

$$
\tan^2\theta_{\mathrm{H}} \mathbb{1} = \Sigma_N^{(\mathrm{H}\,\mathcal{E}_N^{(\mathrm{H})})}, \quad \Sigma_N^{(\mathrm{H}\,\mathcal{E}_N^{(\mathrm{H})})} = R(\phi)\,\mathrm{diag}(\sigma_N, \sigma_N)\,R^{\mathrm{T}}(\phi). \tag{28}
$$

As covariance matrices of ideal and noisy measurements are taken in limit, implying that

$$
\frac{1}{\cos^2\theta} \Sigma_{\text{id}}^{(\mathrm{H})} \sim \Sigma_{\text{id}}^{(\mathrm{H})}, \quad e^{2r} + \text{const} \sim e^{2r}, \tag{29}
$$

it is sufficient to solve for $\cos\theta_{\mathrm{H}}$ and $\sin\theta_{\mathrm{H}}$ from Eq. (28). Doing so yields

$$
\cos\theta_{\mathrm{H}} = \sqrt{\frac{1}{\sigma_x}}, \quad \sin\theta_{\mathrm{H}} = \sqrt{\frac{\sigma_N}{\sigma_x}}. \tag{30}
$$

For double homodyne detection, first, we note that, since $\Sigma_m^{(\mathrm{DH})}$ is anisotropic, as generally quadrature variances $\delta_{1,2}$ are not equal, $\delta_1 \neq \delta_2$, enviroment state must be anisotropic as well. Without loss of generality, we will consider the enviroment to be in pure state with $\det \Sigma_{\text{env}}^{(\text{DH})} = 1$,

$$\Sigma_{\text{env}}^{(\text{DH})} = R(\phi)\,\text{diag}\left(a, \frac{1}{a}\right)\, R^{\text{T}}(\phi), \quad a > 0. \tag{31}$$

Solving Eq. (26) for the case of double homodyne detection, i.e. the system of equations

$$\begin{cases} \delta_1 = \frac{e^{2r}}{\cos^2\theta_{\text{DH}}} + \tan^2\theta_{\text{DH}} a \\ \delta_2 = \frac{e^{-2r}}{\cos^2\theta_{\text{DH}}} + \tan^2\theta_{\text{DH}} a^{-1} \end{cases}, \tag{32}$$

we arrive at the expressions

$$\cos^2\theta_{\text{DH}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)}, \tag{33}$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r} - 1}. \tag{34}$$

2.2 Additive noise channel

We consider additive noise (classical mixing) channel [2] $\oplus_N^{(\gamma)}$, characterized by matrices

$$X^{(\gamma;\,\Phi_N^{(\gamma)})} = \mathbb{1}, \quad Y^{(\gamma;\,\Phi_N^{(\gamma)})} > 0. \tag{35}$$

This is a self-dual channel, $\Phi_N^* = \Phi_N$, that does not affect the first moment and changes the second moment as follows

$$\Phi_N^{(\gamma;\,\Phi_N^{(\gamma)})} : \Sigma \mapsto \Sigma + Y^{(\gamma;\,\Phi_N^{(\gamma)})}. \tag{36}$$

To find matrices $Y^{(\gamma;\,\Phi_N^{(\gamma)})}$ so that

$$\Phi_N^{(\gamma;\,\Phi_N^{(\gamma)})} : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)}, \tag{37}$$

we need to find the difference between matrices of noisy and ideal measurements, i.e.

$$Y^{(\gamma;\,\Phi_N^{(\gamma)})} = \Sigma_m^{(\gamma)} - \Sigma_{\text{id}}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \tag{38}$$

where we defined the covariance matrix of excess noise $\Sigma_N^{(\gamma)}$.

For homodyne detection, from Eq. (7) and (6), we have

$$\Sigma_N^{(\text{H})} = \Sigma_m^{(\text{H})} - \Sigma_{\text{id}}^{(\text{H})} = R(\phi)\text{diag}(\sigma_N, 0)R^{\text{T}}(\phi) \geq 0, \tag{39}$$

where σ_N is given by Eq. (8). While it is possible to define this matrix in the same way as for attenuator channel (see Eq. (28)), it would result in lesser secure key rate (see next section).

For double homodyne detection, from Eq. (15) and (18), we have

$$\Sigma_N^{(\text{DH})}(r) = \Sigma_m^{(\text{DH})} - \Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^{\text{T}}(\phi) \geq 0, \tag{40}$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \tag{41}$$

Note that the elements of $\Sigma_N^{(\text{DH})}$ are dependent on squeezing parameter that lies in the interval (19).

3 GMCS protocol

In the GMCS protocol, Alice sends coherent states $|\alpha = \alpha_1 + i\alpha_2\rangle$ drawn from the Gaussian prior

$$p_A(\alpha) = \frac{1}{\pi V_{\mathrm{A}}} \exp\left[-\frac{|\alpha|^2}{2V_{\mathrm{A}}}\right]. \tag{42}$$

The states pass the composite Gaussian channel $\mathcal{N}_{A\to B} = \mathcal{E}_{T_{\mathrm{ch}}} \circ \mathcal{E}_N$, $N = 1+\xi_{\mathrm{ch}}/(1-T_{\mathrm{ch}})$, where T_{ch} is the channel transmittance and ξ_{ch} is the channel excess noise. Then, the receiver uses either homodyne or double homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the following expressions (see Ref. [1] for derivation)

$$I_{\mathrm{AB}}^{(\mathrm{H})} = \frac{1}{2}\log\left[1 + \frac{4T_{\mathrm{ch}}V_{\mathrm{A}}}{\sigma_x + \xi_{\mathrm{ch}}}\right], \tag{43}$$

$$I_{\mathrm{AB}}^{(\mathrm{DH})} = \frac{1}{2}\sum_i \log\left[1 + \frac{4T_{\mathrm{ch}}V_{\mathrm{A}}}{2\sigma_i + \xi_{\mathrm{ch}}}\right], \tag{44}$$

where σ_x and σ_i given by Eqs. (5) and (14), respectively.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel reads (see Ref. [3])

$$\mathcal{I}_A \otimes \mathcal{E}_{T_{\mathrm{ch}}} \circ \mathcal{E}_N : \quad \Sigma_{\mathrm{TMSV}} = \begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z \\ \sqrt{V^2-1}\sigma_z & V\mathbb{1} \end{pmatrix} \mapsto \Sigma_{\mathrm{AB}} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{45}$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix Σ_{AB} are given by

$$V = 1 + 4V_{\mathrm{A}}, \quad c = \sqrt{T_{\mathrm{ch}}(V^2-1)}, \tag{46}$$

$$V_{\mathrm{B}} = T_{\mathrm{ch}}(V-1) + 1 + \xi_{\mathrm{ch}}. \tag{47}$$

We consider local action of either attenuator channel $\mathcal{E}_N^{(\gamma)}$ (25) or additive noise channel $\Phi_N^{(\gamma)}$ (35), on second mode of TMSVS shared by Alice and Bob (45). In both cases, mutual information is given by $I_{\mathrm{AB}}^{(\gamma)}$ (43), (44), as it is classical.

It follows

$$\mathcal{I}_A \otimes \mathcal{E}_N^{(\gamma)} : \Sigma_{\mathrm{AB}} \mapsto \Sigma_{\mathrm{AB}}^{(\gamma;\,\mathcal{E}_N^{(\gamma)})} = \begin{pmatrix} V\mathbb{1} & c\cos\theta_\gamma\sigma_z \\ c\cos\theta_\gamma\sigma_z & \cos^2\theta_\gamma\left(V_{\mathrm{B}} + \tan^2\theta_\gamma\Sigma_{\mathrm{env}}^{(\gamma)}\right) \end{pmatrix}, \tag{48}$$

$$\mathcal{I}_A \otimes \Phi_N^{(\gamma)} : \Sigma_{\mathrm{AB}} \mapsto \Sigma_{\mathrm{AB}}^{(\gamma;\,\Phi_N^{(\gamma)})} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\mathrm{B}}\mathbb{1} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{49}$$

When the channel $C_\gamma \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat{\rho}_{\mathrm{AB}}^{(\gamma;\,C_\gamma)} = \mathcal{I}_A \otimes C_\gamma(\hat{\rho}_{\mathrm{AB}})$ and the measurements performed by the receiver are assumed to be noiseless. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\mathrm{EB}}^{(\gamma;\,C_\gamma)} = S_{\mathrm{E}} - S_{\mathrm{E|B}} = S_{\mathrm{AB}}^{(\gamma;\,C_\gamma)} - S_{\mathrm{A|B}}^{(\gamma;\,C_\gamma)}, \quad C_\gamma \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \tag{50}$$

gives the Holevo information. The joint entropy

$$S_{\mathrm{AB}}^{(\gamma;C_\gamma)} = g(\nu_1^{(\gamma;C_\gamma)}) + g(\nu_2^{(\gamma;C_\gamma)}), \qquad (51)$$
$$g(\nu) = \frac{\nu+1}{2}\log\frac{\nu+1}{2} - \frac{\nu-1}{2}\log\frac{\nu-1}{2}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (49) given by

$$\nu_{1,2}^{(\gamma;C_\gamma)} = \sqrt{\Delta_{\gamma;C_\gamma}/2 \pm \sqrt{\Delta_{\gamma;C_\gamma}^2/4 - D_{\gamma;C_\gamma}}}, \quad D_{\gamma;C_\gamma} = \det\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}, \qquad (52)$$
$$\Delta_{\gamma;C_\gamma} = \det([\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}]_{11}) + \det([\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}]_{22}) + 2\det([\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}]_{12}), \qquad (53)$$

where $[\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}]_{ii}$ ($[\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{\mathrm{AB}}^{(\gamma;C_\gamma)}$.

Conditional entropies $S_{\mathrm{E|B}} = S_{\mathrm{A|B}}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{\mathrm{A|B}}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det\Sigma_{\mathrm{A|B}}^{(\gamma)}}, \qquad (54)$$
$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)})^{-1}\sigma_z. \qquad (55)$$

Direct computations show that these conditional matrices and therefore their symplectic eigenvalues are independent on channel model. Symplectic eigenvalues are given by

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \qquad (56)$$
$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{\left(V_{\mathrm{B}} + \delta_1\right)\left(V_{\mathrm{B}} + \delta_2\right)}}, \qquad (57)$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [4]

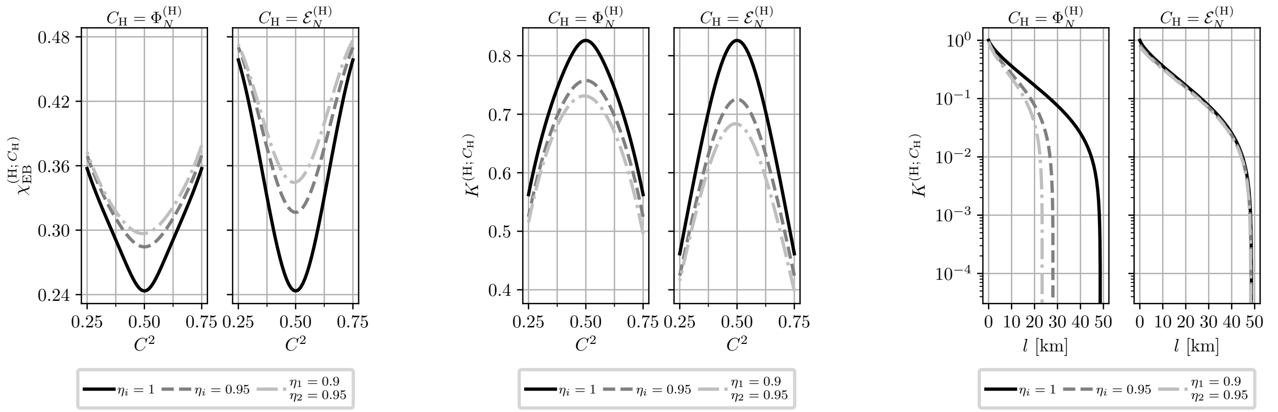
$$K^{(\gamma;C_\gamma)} = \beta I_{AB}^{(\gamma;C_\gamma)} - \chi_{EB}^{(\gamma;C_\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, C_\gamma \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \qquad (58)$$

As symplectic eigenvalues for additive noise channel are calculated per formulas Eq. (52) and do not allow for simplification, in the following sections we focus only on analytic expressions for symplectic eigenvalues for attenuator channel model.

3.1 Homodyne detection

For attenuator channel, from Eq. (48) and (30) we have

$$\Sigma_{\mathrm{AB}}^{(\mathrm{H};\mathcal{E}_N^{(\mathrm{H})})} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}\left(V_{\mathrm{B}} + \sigma_N\right)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \qquad (59)$$
$$\tilde{V}_{\mathrm{B}} \equiv T(V-1) + 1 + \xi, \qquad (60)$$
$$\tilde{c} \equiv \sqrt{T(V^2-1)}, \qquad (61)$$



(a) (b) (c)

Figure 2: (a) Holevo information, $\chi_{\rm EB}^{(\rm H; \mathit{C})}$ and (b) asymptotic secret fraction, $K^{(\rm H; \mathit{C})}$, as a function of the beam splitter transmission, C^2, of the homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\rm ch} = 0.95; \xi_{\rm ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\rm H)}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\rm ch} = 10^{-2}$; losses are 20 dB per 100 km.

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_{\rm B}$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission $T_{\rm ch}$ and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ,

$$T = T_{\rm ch} \cdot \frac{1}{\sigma_x}, \tag{62}$$

$$\xi \equiv \xi_{\rm ch} \cdot \frac{1}{\sigma_x}. \tag{63}$$

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (62) and channel noise (63) by a factor of σ_x^{-1}.

Symplectic eigenvalues of $\Sigma_{\rm AB}^{(\rm H; \mathcal{E}_N)^{(\rm H)}}$, $\nu_{1,2}^{(\rm H; \mathcal{E}_N^{(\rm H)})}$, can be computed using formulas [5]:

$$\nu_{1,2}^{(\rm H; \mathcal{E}_N^{(\rm H)})} = \frac{z \pm [\tilde{V}_{\rm B} - V]}{2}, \quad z = \sqrt{\left(V + \tilde{V}_{\rm B}\right)^2 - 4\tilde{c}^2}, \tag{64}$$

and then substitued in Eq. (51) to calculate joint entropy $S_{\rm AB}^{(\rm H; \mathcal{E}_N^{(\rm H)})}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (50) and asymptotic secret fraction (58) versus beam splitter transmission C^2, as well as secret fraction versus transmission length, for both additive noise (49) and attenuator (48) channels. While channel noise is reduced in this setup (63), so is channel transmission (62), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (43), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive noise channel.

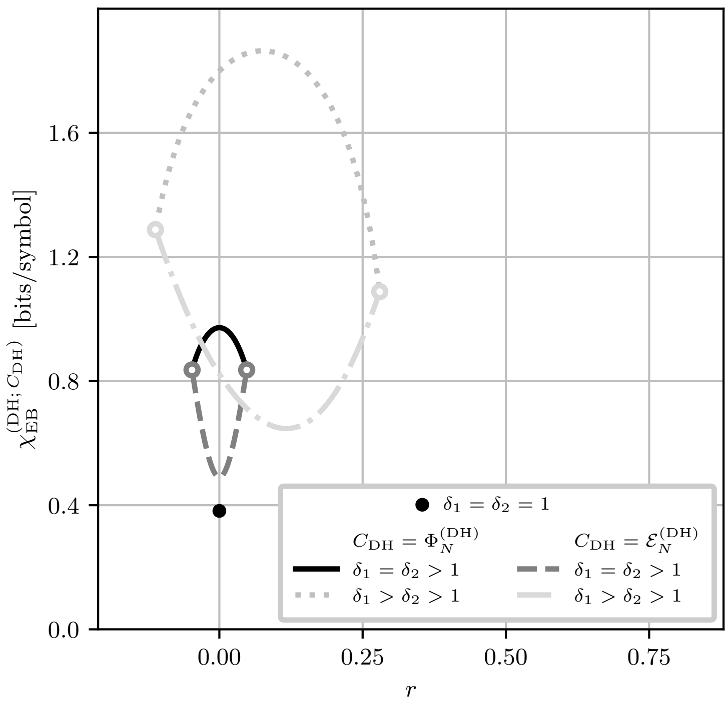


Figure 3: Holevo information, $\chi_{\rm EB}^{(\rm DH;C_{\rm DH})}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dashdotted curves correspond to additive noise channel model, while dashed and dotted curves are for attenuator channel model. Color-matched white-filled dots mark points where Holevo information is undefined at the boundaries of the r interval (19) (see Eq. (34) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{\rm ch} = 0.95$; and $\xi_{\rm ch} = 10^{-2}$.

3.2 Double homodyne

For attenuator channel, from Eq. (48) and (33) we have

$$\Sigma_{\rm AB}^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})} = \begin{pmatrix} V\mathbb{1} & c\cos\theta^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})}\sigma_z \\ c\cos\theta^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})}\sigma_z & B \end{pmatrix}, \tag{65}$$

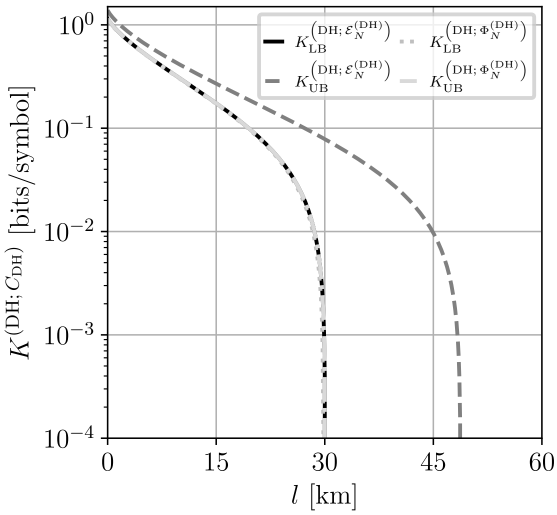
$$B = \cos^2\theta^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})} \begin{pmatrix} V_{\rm B} + \delta_1 - \frac{e^{2r}}{\cos^2\theta^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})}} & 0 \\ 0 & V_{\rm B} + \delta_2 - \frac{e^{-2r}}{\cos^2\theta^{(\rm DH;\,\mathcal{E}_N^{(\rm DH)})}} \end{pmatrix}.$$

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (52). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (19), like in the case of additive noise channel (see Ref. [1]).

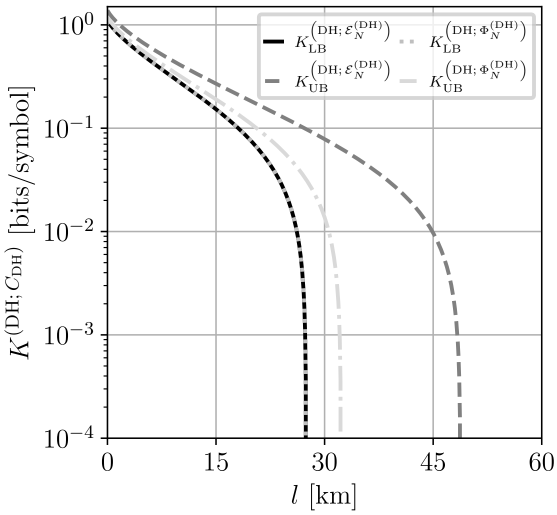
Fig. 3 illustrates dependence of Holevo information on the squeezing parameter, for both additive noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double homodyne with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [1]), with equal values of Holevo information for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of Holevo information on squeezing parameter is symmetrical around $r = 0$ for both channels. For additive noise channel, $r = 0$ is the point of maximal Holevo information, while for attenuator channel $r = 0$ corresponds to minimal Holevo information. Moreover, while Holevo information is unidefined at $r = r_{2,1}$ (see Eq. (33)), the values tend to the same values of joint entropy in the case of additive noise channel, as signified by color-matched white-filled dots in Fig. 3.

In asymmetrical case where $\delta_1 \neq \delta_2$, Holevo information for both channel models become asymmetrical relative to their extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For additive noise channel, Holevo information has a distintive maximum, while for attenuator channel Holevo information has a distintive minimum.



(a) $q = 1$



(b) $q = 1.25$

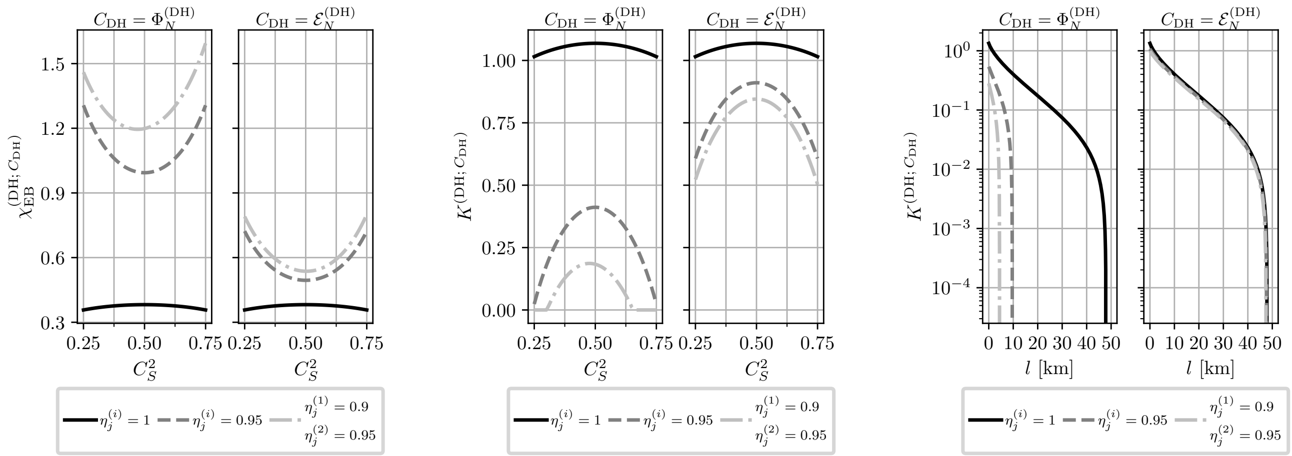
Figure 4: Asymptotic secret fraction $K^{(\mathrm{DH};C_{\mathrm{DH}})}$ versus channel length l, computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(14)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive noise (49) and attenuator (48) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maximum of joint entropy for both models is either at $r = r_1$ or $r = r_2$. Moreover, the upper bound for additive noise channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (50) and asymptotic secret fraction (58) on the signal beam splitter transmission C_S^2, as well as secret fraction versus transmission length if the receiver uses double homodyne detection, for both additive noise (49) and attenuator (48) channels. Unlike the homodyne detection case (see Fig. 2), Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where sharp measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2+r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive noise channel.

Analogously to homodyne detection case (see Fig. 2c), in double homodyne detection case the attenuator channel has greater reach than the additive noise channel, as illustated by Fig. 5c.



(a) (b) (c)

Figure 5: (a) Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{DH});C_{\mathrm{DH}}}$ and (b) asymptotic secret fraction, $K^{(\mathrm{DH};C_{\mathrm{DH}})}$, as a function of the signal beam splitter transmission, C_S^2, of the double homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\mathrm{ch}} = 0.95; \xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\mathrm{DH};C_{\mathrm{DH}})}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel models $r = \frac{r_2+r_1}{2}$ (see main text).

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